

CLAUDE SKILLS - SETUP & INSTALLATION

PE ESG Analysis Skills

A reusable skill set that produces ESG analyses of private equity targets, from a one-line verdict to a full structured report.

Installation guide for deployment in your own organisation
Includes the four skill templates, a self-contained Word export module, light/full modes, optional modules, and two reference appendices

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1. Overview

This guide lets you install, in your own organisation, a set of Claude skills that produce institutional-grade ESG analyses of **private equity targets**. The skills work from the sources **you** provide for a given deal, in any combination: a document-store link (SharePoint, Google Drive, Box, a data room), an information memorandum (IM / CIM), an internal investment note or memo, a company name, or a website URL.

The system delivers the same analysis at three depths:

Format	Output	Typical use
Ultra-short	One or two sentences, only the problematic points (or "limited ESG issues")	Rapid screening of inbound deal flow

Short	One-pager, under 2000 characters (Products / ESG risks / Governance)	Opportunity sheet, early committee
Long	Full structured report, auto-generated as a styled Word (synthesis table + colour-coded blocks)	Commitment committee, investment memo

In every format, the full analysis runs (sources, website, controversy research on the company and its executives, employee and client reviews, SASB sector checklist). The ultra-short format only shortens the *output*, not the work behind it.

2. The skill set and naming convention

You install one main skill and three shortcut skills. Each shortcut simply forces one output format and calls the main skill, so the method lives in one place.

Skill (default name)	Role
<code>pe-esg-analysis</code>	Main skill. Holds the full method. Asks which format to use.
<code>pe-esg-analysis-ultra-short</code>	Shortcut: ultra-short verdict, no format question.
<code>pe-esg-analysis-short</code>	Shortcut: one-pager, no format question.
<code>pe-esg-analysis-long</code>	Shortcut: full report, no format question.

These default names work as-is. **Optionally**, you can prefix them with your firm or fund name to disambiguate, for example `acme-pe-esg-analysis`, `acme-pe-esg-analysis-short`, etc. If you add a prefix, apply it to all four folder names and update the `../pe-esg-analysis/` path inside each shortcut skill so it still points to the main skill.

Why "pe" in the name? The "pe" (private equity) tag signals that these analyses target **private, non-listed SMEs**, where materiality, data availability and the ESG approach differ markedly from large listed companies. In a dedicated PE firm this is implicit; in a broader organisation it may not be, so the tag keeps the scope unambiguous.

Naming rule (important). A skill's **name** and **description** must never contain angle brackets (the < and > characters), whether or not you add a company name. Angle brackets are read as XML tags and the skill is rejected with "cannot contain XML tags". This is why the default names use no placeholder. Each SKILL.md **description** must also be at most **1024 characters**. Angle-bracket placeholders inside the body of a SKILL.md (e.g. `<Company name>` in the Markdown contract) are fine; only the name and description fields are validated.

3. Requirements

- Claude with the **skills capability** enabled (e.g. Claude desktop Cowork mode, or Claude Code).
- Access to the documents you want analysed: either upload them per deal, or connect your document store (SharePoint, Google Drive, Box, etc.) so Claude can read from a link you provide.
- Web access (for website exploration, controversy research and employee/client reviews).

4. Installation

Option A - Guided setup (recommended)

Open Claude with skills enabled and paste the following setup prompt. Claude will ask the two install-time questions (section 5), then create the four skills.

```
I want to install the "PE ESG Analysis" skill set from the installation guide.
First, collect the configuration by asking me with the interactive
multiple-choice question interface (clickable options) - NOT as a plain-text
list - grouping the questions into a single set I can click through:
(1) The name of my asset-management company / fund. Options: let me type it, or
    "Skip". Do NOT explain in the question that this name will be used to name the
    skills; simply ask for the firm name. (Internally: if I give a name, use it as
    the skill-name prefix; if I skip, use the default names with no prefix.)
(2) Default analysis language. Options: "Match each deal's source documents",
    "English", "French", or let me type another.
(3) Analysis mode. Options: "Light (token-optimised, recommended for free or
    small plans)" or "Full (thorough, default)".
(4) Optional modules to enable (long format only; multiple choice). Options:
    "competitor-benchmark", "exclusion-screening", "double-materiality", "none".
If I choose exclusion-screening, then in a follow-up ask me for my exclusion list,
telling me it is optional and that if I do not provide one, the tool will screen
against the standard exclusions commonly used by private-equity investors.
Then create these four skills (apply my prefix if I gave one, otherwise use the
names as-is), set my language choice in the main skill's "Output language" rule,
and fill the main skill's "Installation configuration" block with my mode and
enabled modules:
    pe-esg-analysis          (main, full text below)
    pe-esg-analysis-ultra-short
    pe-esg-analysis-short
    pe-esg-analysis-long
Also add the reference files:
    pe-esg-analysis/references/impact-themes.md      (Appendix A)
    pe-esg-analysis/references/sasb-disclosure-topics.md (Appendix B)
    pe-esg-analysis/references/exclusion-list.md      (only if I enabled exclusion-screening; use my
list)
And add the Word export script:
    pe-esg-analysis/scripts/render_esg_long.py      (section 8)
Important: never put angle brackets (the < and > characters) in a skill name
or description, with or without a prefix, and keep each description at most 1024
characters, or saving the skill will fail with "cannot contain XML tags".
Use the skill texts and both reference lists from the installation guide verbatim.
```

Option B - Manual install

1. Create one folder per skill, named exactly as in section 2. The default names are ready to use; only add a company prefix if you want one (and then update the `../pe-esg-analysis/` path in each shortcut).
2. In each folder, create a `SKILL.md` and paste the matching template from section 7. The templates already use the default names and save as-is - do not add angle brackets to the name or description.
3. In the main skill folder, create `references/impact-themes.md` (paste Appendix A), `references/sasb-disclosure-topics.md` (paste Appendix B), and `scripts/render_esg_long.py` (paste section 8). Set the main skill's **Installation configuration** block (analysis mode, enabled modules). If you enable exclusion screening, also create `references/exclusion-list.md` with your own exclusion list.
4. Install the skills the way your Claude environment expects (for example, add them under Settings > Capabilities, or place the folders in your skills directory). Zipping a skill folder as a `.skill` archive makes it installable in one click in Cowork.

If a save is rejected with "SKILL.md description cannot contain XML tags" or "field 'name' ... cannot contain XML tags", a < or > character is present in the name or description - remove it (this is what happens if a placeholder like the old <company> is left in). Also keep each description under 1024 characters.

5. Install-time questions

During setup, the installer asks all of these settings at once through a clickable multiple-choice interface (fund name, language, mode and optional modules together):

- **Company / fund name (optional).** If provided, it prefixes the skill names (e.g. `acme-pe-esg-analysis`). If skipped, the default names are used (`pe-esg-analysis` ...). Never use angle brackets in the name, with or without a prefix.
- **Default analysis language.** The language analyses are written in by default. If you choose "match the source documents", each analysis is written in the language of the deal documents (internal note first, then the IM), falling back to English. Whatever you choose is written into the main skill's *Output language* rule.
- **Analysis mode: light or full.** Light mode is token-optimised (recommended for free or small plans): it reads the IM via a sub-agent that returns only an ESG extract, pulls just the target's SASB sector block, and caps web research; full mode is the thorough default. The choice is written into the main skill's configuration block.
- **Optional modules (long format only, multi-select).** Add any of: a competitor ESG benchmark appendix; an exclusion-list screening appendix; a double-materiality (SASB) appendix. If you enable exclusion screening, you may provide your exclusion list at install (optional) - it is saved as `references/exclusion-list.md`; if you do not provide one, the tool screens against the standard exclusions commonly used by private-equity investors. Each module is rendered as a table in the Word report.

6. How the analysis behaves

The full behaviour is specified inside the skill templates (section 7). Key points:

- **User-provided sources.** The skill gathers from whatever you supply for the deal (a store link, an IM/CIM, an internal note, a name, a website). It does not assume any fixed folder layout.
- **Two mandatory check-ins.** The skill pauses to ask you when (1) the **company name or website is ambiguous** (common name, several legal entities, unclear target), and (2) **no information memorandum** was provided or it cannot be found in the store you gave - in which case it asks you to confirm that no IM exists (or to provide it) rather than silently proceeding.
- **Depth regardless of format.** Website ESG pages, controversy research on the company and named executives, employee and client reviews, and the SASB sector checklist run in all formats.
- **SASB as a safety net.** The skill reads only the target's sector block from the bundled SASB reference (Appendix B) to verify no material issue was missed; it may reuse SASB wording when relevant but keeps your own report structure.

- **Impact themes.** When the products or services have a positive impact, the skill names the theme(s) from a fixed Impact themes list (Appendix A), using only the theme names; the sub-theme examples are selection guidance and never appear in the output.
- **Adaptive section name.** The products/services section and its synthesis row are named "Impacts of products", "Impacts of services" or "Impacts of products and services" depending on what the company mainly sells, and the synthesis row is suffixed with the main impact theme.
- **Analysis mode (light vs full).** Set at install. Light mode cuts token consumption (IM read via a sub-agent extract, SASB sector block only, capped web research) at the cost of some depth; full mode is thorough. See the configuration block and the "Analysis mode" section in the main skill template (7.1).
- **Optional modules (long format only).** If enabled at install, the long report gains appendices - a competitor ESG benchmark, an exclusion-list screening, and/or a double-materiality (SASB) matrix - each rendered as a table by the Word script.

7. Skill templates

Create each file below as `SKILL.md` in its own folder. The templates use the default names and save as-is; optionally prefix the folder names with your firm name (and update the `../pe-esg-analysis/` path in the shortcuts). Keep each description under 1024 characters and free of angle brackets.

7.1 pe-esg-analysis/SKILL.md (main)

```
---
name: pe-esg-analysis
description: Produces an ESG analysis of a private equity target company in one of three formats (ultra-short verdict, short one-pager, or long structured report). On launch it asks which format is wanted, unless a format was already imposed by a shortcut skill (pe-esg-analysis-ultra-short / -short / -long). Use whenever the user asks for an ESG (or CSR/RSE) analysis, an ESG screening, an ESG one-pager, an ESG report, or an ESG opinion for an investment committee, or provides a company name (and possibly a website, a memo, or a document-store link) expecting an ESG opinion, even without using the word ESG.
---

# PE ESG Analysis (ultra-short, short one-pager, or long report)

This skill produces an ESG analysis of a private equity target company, for investment committees and deal memos. It exists in three formats that share the same source collection and method but differ in output length:
- Ultra-short: a one or two sentence verdict.
- Short: a one-pager (three dense paragraphs, under 2000 characters) plus a complementary synthesis.
- Long: a full structured report (three sections with subsections and financial impacts) plus a complementary synthesis.

## Installation configuration
Set at install time; leave the defaults if unsure. These settings are baked into this file when the skill is installed.
- Analysis mode: full (choose "light" or "full"; light = token-optimised, recommended for free or small plans)
- Default analysis language: match source documents (or a specific language)
- Optional modules (long format only): none (any of: competitor-benchmark, exclusion-screening, double-materiality)
- Exclusion list: references/exclusion-list.md (optional; used if exclusion-screening is enabled and this file exists; otherwise a standard PE exclusion set is applied)

Installer note: to keep this file lean, you may delete the analysis-mode branch you did not choose and the optional-module descriptions you did not enable.

## Step 0: determine the format
- If a format was already imposed by the call (e.g. invoked from pe-esg-analysis-ultra-short / -short / -long, or the user already said "ultra-short"/"verdict", "short"/"one-pager", or "long"/"full report"), do not ask: keep that format and go straight to source collection.
- Otherwise, ask the user a single short multiple-choice question for the format. Use the same interaction to confirm, if needed, the company name and the source(s) to use.

## Inputs and source collection
The only indispensable input is the target company name. Everything else is supplied by the user and/or actively gathered by Claude from the sources the user provides. Accept any combination of:
- a document-store link (SharePoint, Google Drive, Box, a data room, etc.),
- an information memorandum (IM) / confidential information memorandum (CIM) / "info memo" (the same document),
- an internal investment note or memo,
- a company name,
- a website URL.

Before launching, briefly confirm with the user (ideally in the same interaction as the format choice): the target company name, and which source(s) to use (a link, an uploaded document, or a search by company name). Do not re-ask for information that is accessible in the sources provided.

Then actively gather the sources. Distinguish two families:
1. External deal documentation (from the sell-side / M&A advisor). The IM/CIM is central: it is the factual reference on the company (activity, product mix, clients, geography, headcount, executives, ownership). Read it in full, even if large (by chunks or via a dedicated sub-agent). If text extraction fails or is partial (flattened graphical PDF, truncated export), do not assume it has been read: check
```

coverage and complete missing sections via a page-by-page visual read in a sub-agent; if still impossible, ask the user to provide the IM rather than silently proceeding, and flag the gap in Sources. Also look for the teaser, vendor due diligence (financial / strategic / ESG), management presentation, Q&A and due-diligence reports.

2. Internal investment documents. These are the source of the internal reading (thesis, view on management quality and governance, watch points) and of the tone, language and committee stage. Look for the internal opportunity sheet / one-pager and the investment committee notes.

Read both families. Never conclude on the business from internal documents alone, nor on governance and management quality from the IM alone.

Identify the company website from the IM or internal note; failing that, via web search. Complete with the public research described in the method below.

Mandatory user check-ins

Two situations require asking the user before continuing (one short question each):

1. Ambiguous company name or website. If the company name is common, maps to several legal entities, or it is unclear which website/entity is the target, ask the user to confirm the exact name and website before proceeding. Never analyse the wrong entity.

2. No information memorandum. If no IM was provided, or it cannot be found in the document store the user supplied, explicitly ask the user to confirm that no IM exists (or to provide it) before continuing. Do not silently proceed as if the IM were simply absent: its absence materially changes the analysis and must be acknowledged and recorded in Sources.

Documentary prerequisites

Before producing the analysis, check that the indispensable documents are present. If one is missing, do not launch: warn the user in one or two sentences (stating which document is missing and where it was searched) and ask whether to continue anyway or provide it.

- IM (all formats): indispensable in all cases. If unavailable and the user cannot provide it, do not launch without the user's explicit agreement to proceed anyway.

- Short format: the internal one-pager (if the user's process produces one) is recommended.

- Long format: the internal one-pager and the first investment committee note are recommended.

- Ultra-short format: only the IM is indispensable; internal notes are useful but not blocking (this format is meant for rapid screening).

If the user agrees to proceed despite a missing document, do so and flag the gap explicitly in the Sources section.

Output language

If a default analysis language was configured at installation, write the analysis in that language. Otherwise, write in the same language as the internal investment documents if any; failing that, the language of the IM; failing that, English.

Method (common to all formats)

1. Retrieve and read the provided sources: external deal documentation (IM first) for the factual base, and internal investment documents for the thesis, governance and management quality.

2. Systematically explore the company website for ESG content. Do not rely on the home page or a generic search. Fetch the typical pages: About / Company; Sustainability / CSR / Commitments / Impact; Quality / Certifications / QHSE; Careers; Legal information (footer often lists certification logos); Timeline / Key dates; flagship product pages (sometimes quantified impact data); Press / News. This exploration is mandatory before concluding that ESG content, a formalised CSR policy, or certifications are absent.

3. Search for controversies via web search on two axes: the company and its sector; and the named executives identified in the IM or internal notes (at minimum founder(s), CEO / Chair, main operating shareholders).

4. Systematically check employee reviews (Glassdoor, Indeed, etc.) and client reviews (Trustpilot, Google Reviews, sector forums). Mandatory even if nothing notable comes out. When the clientele is exclusively institutional / B2B / B2G, note that consumer review platforms are not relevant and read satisfaction from other signals (tender win rate, framework agreements, renewals). Browser fallback: many review platforms block direct fetching (anti-scraping); when a fetch returns empty while search shows reviews exist, switch to a rendered-browser read; if access remains impossible, say so rather than pretending to rely on a source you could not read.

5. If the sector is known for recurring controversies, run dedicated research on documented sector incidents (dates, places, magnitudes, operator involved, fines).

6. Load the SASB sector checklist: read the bundled file references/sasb-disclosure-topics.md, locate the target's sector (search by industry name) and read only that block. SASB is the best available exercise for identifying sector issues: use it to verify that no material issue has been forgotten, and reuse an issue's wording as-is when relevant. This is not mandatory: some SASB issues are very relevant to the target, others less so; adapt, merge or drop case by case. Keep your own report structure and format.

7. Identify the investment committee stage (early screening vs commitment) and calibrate the recommendation accordingly.

8. Write the analysis strictly in the format chosen at Step 0.

Analysis mode (light vs full)

Apply the mode set in the installation configuration.

- Full mode: run the method above exactly as described (thorough). This is more token-intensive.

- Light mode (token-optimised, for free or small plans): reduce consumption as follows, accepting less depth, while keeping the same output structure:

- IM: read it via a dedicated sub-agent that returns only a tight ESG extract (activity, product/revenue mix, clients, geography, headcount, executives, ownership, and any ESG / HSE / governance mentions); prefer text extraction; use page-by-page image reads only for the few sections where text extraction fails. Do not load the full IM into the main context.
- SASB: extract only the target's sector block from references/sasb-disclosure-topics.md with a targeted search (grep), never read the whole file into context.
- Web research: limit controversy searches to the one or two key executives; fetch only the core website pages (About, Sustainability, Quality); prefer search snippets over full-page fetches; use a rendered browser only when a direct fetch fails; skip the dedicated sector-incident deep dive.

Committee-stage calibration

- Early screening committee: do not lay out ESG improvement roadmaps or value-creation levers. Flag gaps factually ("not communicated at this stage").
- Later / commitment committee: improvement areas, action plans and value-creation levers become relevant.

Output formats

Apply only the format chosen at Step 0.

- Ultra-short: deliver the verdict (one or two sentences) in the conversation; no file.
- Short: deliver the one-pager text in the conversation; no file unless explicitly asked.
- Long: generate a Word report with the bundled script scripts/render_esg_long.py (self-contained: it builds the .docx from scratch, no external template and no logo), then present the .docx. See "Long-format Word output" below for the Markdown contract and the command.

> ULTRA-SHORT FORMAT (one or two sentence ESG verdict)

Produces a very tight ESG verdict: one or two sentences, keeping only the potentially problematic points (exposure to an excluded activity, a poorly managed material environmental or social risk, a confirmed controversy on the company or an executive, concerning governance). This is a rapid screening format.

Run the full method anyway (sources, website, controversies on company and executives, employee and client reviews, SASB checklist): the output is short, but the verdict must rest on the same analytical depth as the other formats. Do not cut corners on research because the output is brief.

Rules:

- One or two sentences maximum, no title, no bullets, no table.
- Mention only what is potentially problematic. If there is little to flag, say so simply, e.g.: "Limited ESG issues: B2B services activity with no particular material exposure, no controversy identified on the company or its executives."
- When there are watch points, state them directly and factually, from most to least material, without listing mitigations or improvement areas. If a point rests on a controversy, give its nature (and date) in a few words.
- Stay factual and calibrated to the committee stage.

The complementary synthesis, the Sources section, the governance-quality rating and any report file do not apply to the ultra-short format.

> SHORT FORMAT (one-pager, under 2000 characters)

Three dense paragraphs, no titles, in this order:

Paragraph 1, Products and services (title the angle "products", "services" or "products and services" depending on what the company mainly sells). How the products / services address environmental or social issues, or conversely raise problems; quantify the share of revenue exposed when available. Enter directly through the ESG angle without re-describing the activity. When the activity has a positive impact, name the relevant impact theme(s) from the Impact themes list (Appendix A), using only the theme names, not the sub-theme examples. If impacts are limited, say so in one sentence and move on; do not look for indirect angles that read as filler.

Paragraph 2, ESG risks. Main environmental and social risks given the sector and model, and how the company responds (certifications, formalised policies, dedicated team, published indicators). Aim for at least three risks for an industrial or exposed company, numbered (i), (ii), (iii). Do not confuse sustainability risks with the business model's financial risks. For each risk, state explicitly whether online research surfaced a controversy.

Paragraph 3, Governance. Ownership, management stability, upcoming changes, specific risks. Prefer people-centred elements over structures. Reuse any internal material on management quality. Close with the governance quality rating: "Governance quality: <level>" (scale below); this line is outside the 2000-character count.

Total body under 2000 characters (excluding sources and complementary synthesis). No titles or subtitles in the body.

```

## > LONG FORMAT (full ESG report)
Opens with a synthesis, then three main sections with Markdown titles, each with subsections.

# Synthesis: up to three paragraphs giving an overall ESG reading, readable on its own; do not re-describe
the activity; do not open with a global ESG grade; qualify issue by issue; one sentence on governance; one
sentence on the controversy review result. The synthesis table is generated automatically; its products/
services row shows the section name followed by the main impact theme, e.g. "Impacts of services:
Education & knowledge".

# Impacts of products and services: name this section "Impacts of products" if the company mainly sells
products, "Impacts of services" if it mainly sells services, or "Impacts of products and services" if
both. How products / services address or harm environmental and social issues; quantify exposed revenue
and a sourced order of magnitude of impact when relevant. Pick the relevant impact theme(s) from the
Impact themes list (references/impact-themes.md, Appendix A), using only the theme names; never reproduce
the sub-theme examples, which are there only to guide your selection. Mandatory subsection ## Potential
financial impacts, followed by its level (Strong / Medium / Weak) and a sign (+ / = / -) when Strong or
Medium.

# Main environmental and social risks: at most five subsections, one per main risk, each titled ## with
(Exposure: Strong / Medium / Weak | Maturity: Very advanced / Advanced / Medium / Limited / Poor
practices / ?). For each risk: a first paragraph on why it is a risk for this specific company; one or
more paragraphs on what the company does about it; mandatory subsection ### Potential financial impacts
(level plus sign). Exposure to exclusion policies is never a risk in this section.

# Governance (Governance quality: <level>): ## Economic governance (with ### Potential financial impacts),
## CSR governance (no financial-impact subsection), and ## Business ethics if a Medium or Strong ethics/
compliance issue exists (with ### Potential financial impacts). Business ethics, when included, is titled
exactly "Business ethics" and carries both Exposure and Maturity in its title bar (like a risk), not
Maturity alone.

Notations: each risk carries (Exposure | Maturity); each financial-impact subsection carries its level
(and + / = / - if Strong or Medium); the governance section title carries the governance quality rating.

### Long-format Word output (self-contained script)
Deliver the long report as a Word file produced by scripts/render_esg_long.py. The script is self-
contained: it builds the .docx from scratch (synthesis table, coloured title blocks, pillar pastilles),
with no external template and no logo. Do not hand-craft the Word or write XML: write the analysis to a
Markdown file following the contract below, then run the script.

Markdown contract:

TITLE: <Company name>
SUBTITLE: Full ESG report | <committee stage> | <month year>

# Synthesis
<synthesis paragraphs>          (the synthesis table is inserted AUTOMATICALLY; do not write it)

# Impacts of products and services      (rename to "Impacts of products" or "Impacts of services" per
scope)
[HEADER] scope=<products|services|both> | note=<Strong|Medium|Weak> | pillar=<E|S|E/S> | exposure=<Direct|
Indirect> | themes=<main theme first, names from the Impact themes list>
<paragraphs>
### Potential financial impacts (Medium, +)
<paragraphs>

# Main environmental and social risks
## [S] <Risk title> (Exposure: Strong | Maturity: Advanced)
<paragraphs>
### Potential financial impacts (Medium, -)
<paragraphs>
## [E] <Risk title> (Exposure: Medium | Maturity: Limited)
... (up to 5 risks; each title prefixed with the pillar [E], [S] or [E/S])

# Governance
## [G] Economic governance (Maturity: Medium)
<paragraphs>
### Potential financial impacts (Medium, +)
<paragraphs>
## [G] CSR governance (Maturity: Limited)
<paragraphs>
## [G] Business ethics (Exposure: Medium | Maturity: Medium)    (include only if a Medium/Strong ethics or
compliance issue exists)
<paragraphs>
### Potential financial impacts (Medium, =)
<paragraphs>

```

```
# Complementary synthesis
## Employee reviews
## Client reviews
## Online controversy review
## Controversy review on executives
```

```
# Sources
<paragraphs>
```

(If optional modules are enabled, add their appendix sections here, after Sources, each as a "# Appendix - ..." heading followed by a Markdown table; see "Optional modules".)

Contract rules:

- The tags [E] / [S] / [E/S] / [G], the [HEADER] line and the (Maturity: ...) / (Exposure: ... | Maturity: ...) notations are consumed by the script and do not appear as such in the Word: they drive the pastilles, the colours and the synthesis table.
- Governance: in the Word, the governance rating is not global in the # Governance title; each subsection carries it via ## [G] ... (Maturity: <level>).
- Business ethics: include this [G] subsection only when a Medium or Strong ethics/compliance issue exists; title it exactly "Business ethics" and give it both Exposure and Maturity, e.g. ## [G] Business ethics (Exposure: Medium | Maturity: Medium), so its title bar shows exposure like a risk.
- Synthesis table: generated automatically from the pillars, ratings and "Potential financial impacts" of each block. The overall arrow ++ / + / = / - / -- combines Exposure and Maturity (default matrix); the \$ marker follows the financial-impact sign (green if +, orange if -). Override the matrix when needed with a "| Synthesis: <arrow>" token, e.g. (Exposure: Strong | Maturity: Medium | Synthesis: -).
- scope drives the section label in the synthesis table; the synthesis row shows that label followed by ":" and the main impact theme (the first theme listed in themes). Use theme names from the Impact themes list only; never the sub-theme examples.
- Plain paragraphs stay plain text.

Command:

```
python3 scripts/render_esg_long.py --content analysis.md --out "ESG Analysis - CompanyName.docx"
```

Then present the produced .docx.

```
---
```

Optional modules (long format only)

Produce a module only if it is listed in the installation configuration, and only in the long format (never in ultra-short or short). Each module is an appendix at the end of the report, after Sources, rendered as a table using the Markdown-table syntax below (the Word script turns it into a bordered table). Keep modules factual; never fabricate ratings or facts.

- competitor-benchmark -> "# Appendix - Competitor ESG benchmark". Identify the target's main competitors automatically (from the IM and the sector), unless the user specifies particular competitors. Table columns: Company | Certifications / ratings | ESG commitments | Controversies. Put the target on the first data row, then the competitors.
- exclusion-screening -> "# Appendix - Exclusion screening". Screen the target against every criterion in references/exclusion-list.md if that file is present. If it is absent (the user did not provide a list), screen instead against the standard exclusions commonly used by private-equity investors: controversial or prohibited weapons, tobacco production, thermal coal and coal-fired power generation, oil and gas extraction, gambling, adult entertainment, predatory lending, and activities in breach of the UN Global Compact principles. Table columns: Exclusion criterion | Status | Note, where Status is Compliant, Exposed, or To verify.
- double-materiality -> "# Appendix - Double materiality (SASB)". Over the target's SASB sector topics, assess two axes. Table columns: SASB topic | Impact materiality | Financial materiality | Comment, where each materiality is High, Medium or Low (impact = the company's effect on the environment and society; financial = the effect of the issue on the company).

Markdown-table syntax (header row, separator row, then data rows):

```
| Column A | Column B | Column C |
| --- | --- | --- |
| value | value | value |
```

Section naming and impact themes (common)

Name the products/services impacts section, and its row in the synthesis table, "Impacts of products" if the company mainly sells products, "Impacts of services" if it mainly sells services, or "Impacts of products and services" if both. Choose the impact theme(s) from the Impact themes list (references/impact-themes.md, Appendix A of the installation guide). Use only the theme names in the analysis; the sub-theme examples in that list are there solely to guide your theme selection and must never appear in the output. In the synthesis table, the products/services row is followed by the main impact theme, e.g. "Impacts of services: Education & knowledge".

Taking a position on financial impacts (common)

In each "potential financial impacts" subsection, take a position: cite positives and negatives but conclude one way or the other, or state explicitly that they balance out. Avoid the "Positively... Negatively..." structure. The level and sign must be consistent with the position taken.

Governance quality rating (common, not for ultra-short)

Always provide an overall governance-quality rating on the scale below (use ? rather than inventing a rating):

- Very advanced: exemplary, formalised governance (structured and active bodies, strong alignment of executives, prepared succession, robust compliance and internal control, no blind spots).
- Advanced: solid governance, aligned and experienced executives, bodies in place, minor and controlled watch points.
- Medium: functional but improvable (key-person dependence, succession not yet prepared, partial formalisation, some related parties to frame).
- Limited: loosely structured governance, significant dependencies or blind spots, low formalisation, missing information.
- Poor practices: confirmed negative signals (unaddressed conflicts of interest, opacity, executives' judicial or disciplinary records, governance litigation).
- ?: insufficient information to rate (flag the missing elements).

Certifications glossary (common)

Always state what each cited certification is for and attach it to the pillar it actually addresses, e.g. ISO 9001 (quality), ISO 14001 (environment), ISO 45001 (occupational health and safety), ISO 50001 (energy management), ISO 27001 (information security), EcoVadis (supplier CSR rating, Bronze / Silver / Gold / Platinum), B Corp (certified mission-driven company). Distinguish certifications from certifying bodies (e.g. Bureau Veritas, SGS, TUV, Apave, Dekra certify; they are not certifications). Explain every sector acronym and in-house product or tool name at first use.

Complementary synthesis (short and long formats only)

After the main body, produce four short, distinct syntheses with dedicated titles, mandatory even when research found nothing notable (not for ultra-short): 1. Employee reviews; 2. Client reviews; 3. Online controversy review (company); 4. Controversy review on executives (named). Keep each short. For each review block, report the overall rating and review volume, recurring themes, and any alert signals; do not detail sub-category scores.

Sources (short and long formats only)

List the sources consulted in two categories: information provided by the company in the process (state the nature of the information, without naming each internal document individually; if the document store could not be located or used, say so); and external sources, named (website with the specific pages visited, company registries, press, court decisions, web searches with main keywords).

Writing constraints and pitfalls

- Deliver per format (ultra-short / short: text in the conversation; long: written report).
- Factual, measured, analytical tone; no activist jargon or greenwashing. Never invent figures (headcount, revenue, % of revenue on a theme, emissions, online ratings).
- Do not name internal documents in the body; present facts directly; for missing information write "not communicated to us at this stage".
- Do not make exposure to exclusion policies a risk in the "Main environmental and social risks" section; treat it, only if material, under the products/services impacts section.
- Do not confuse sustainability risks with the business model's financial risks.
- Do not confuse indirect sector exposure with direct impact; characterise the real degree of operational involvement (platform / network / franchise / intermediation models).
- Never claim "no ISO certification" / "no formalised CSR policy" / "no carbon footprint" without first fetching the site's About / Company / Sustainability / Quality / Certifications / Legal information / Timeline pages.
- Do not list acronyms (certifications or sector terms) without a short gloss at first use.
- Do not omit the complementary synthesis (short and long), even when all research is inconclusive.

7.2 pe-esg-analysis-ultra-short/SKILL.md

```
---
name: pe-esg-analysis-ultra-short
description: Writes an ultra-short ESG verdict (one or two sentences) keeping only the potentially
problematic points of a private equity target, or stating that ESG issues are limited if there is nothing
notable. Shortcut that runs pe-esg-analysis directly in ultra-short format, without asking the user for
the format, while still running the full analysis. Use when the user asks for an express ESG opinion, a
one-line ESG verdict, an ESG flash, a very fast ESG screening, or an ultra-short ESG analysis.
---

# PE ESG Analysis Ultra Short (shortcut to pe-esg-analysis, ultra-short format)

This skill is a shortcut to pe-esg-analysis, with the ultra-short format (one or two sentence ESG verdict)
already imposed.

Procedure:
1. Read ../pe-esg-analysis/SKILL.md to load the full method (source collection, website exploration,
controversy research on the company and executives, employee and client reviews, SASB sector checklist).
2. Run the full method (collection and research steps identical to the other formats), then apply only the
"> ULTRA-SHORT FORMAT" section for the output: one or two sentences keeping only the potentially
problematic points, or the statement that ESG issues are limited if there is nothing notable. The output
is very short, but the underlying analysis stays complete.
3. Do not run Step 0 (format choice) and do not ask the user which format they want: the format is fixed
to ultra-short. Go straight to confirming the company name and the source(s), then to source collection.
4. For the ultra-short format, the complementary synthesis, the Sources section, the governance-quality
rating and any report file do not apply (see the "> ULTRA-SHORT FORMAT" section in pe-esg-analysis).

Everything else (inputs, method, mandatory user check-ins, SASB checklist, pitfalls) is defined in pe-esg-
analysis and must be followed as-is.

Note: if you renamed the skills with a company prefix at install time (e.g. acme-pe-esg-analysis), use
that same prefix in the path at step 1 (../acme-pe-esg-analysis/SKILL.md).
```

7.3 pe-esg-analysis-short/SKILL.md

```
---
name: pe-esg-analysis-short
description: Writes a concise ESG analysis in one-pager format (Products / ESG risks / Governance, under
2000 characters) for a private equity target. Shortcut that runs pe-esg-analysis directly in short format,
without asking the user for the format. Use when the user explicitly asks for a short ESG analysis, an ESG
one-pager, an ESG screening, or a concise ESG opinion for an investment committee.
---

# PE ESG Analysis Short (shortcut to pe-esg-analysis, short format)

This skill is a shortcut to pe-esg-analysis, with the short format (one-pager, under 2000 characters)
already imposed.

Procedure:
1. Read ../pe-esg-analysis/SKILL.md to load the full method (source collection, website exploration,
controversy research, employee and client reviews, SASB sector checklist, complementary synthesis).
2. Run that method in short format only: apply the "> SHORT FORMAT" section and its constraints, plus the
complementary synthesis and the sources.
3. Do not run Step 0 (format choice) and do not ask the user which format they want: the format is fixed
to short. Go straight to confirming the company name and the source(s), then to source collection.

Everything else (inputs, method, mandatory user check-ins, certifications glossary, complementary
synthesis, sources, constraints, pitfalls) is defined in pe-esg-analysis and must be followed as-is.

Note: if you renamed the skills with a company prefix at install time (e.g. acme-pe-esg-analysis), use
that same prefix in the path at step 1 (../acme-pe-esg-analysis/SKILL.md).
```

7.4 pe-esg-analysis-long/SKILL.md

```
---
name: pe-esg-analysis-long
description: Writes a long, structured ESG report for a private equity target, with three main sections
(Impacts of products and services / Main environmental and social risks / Governance), financial-impact
subsections, and a mandatory complementary synthesis. Shortcut that runs pe-esg-analysis directly in long
```

format, without asking the user for the format. Use when the user explicitly asks for a detailed ESG analysis, a long ESG analysis, a full ESG report, or an ESG analysis for an in-depth investment memo.

PE ESG Analysis Long (shortcut to pe-esg-analysis, long format)

This skill is a shortcut to pe-esg-analysis, with the long format (full structured report) already imposed.

Procedure:

1. Read ../pe-esg-analysis/SKILL.md to load the full method (source collection, website exploration, controversy research, employee and client reviews, SASB sector checklist, complementary synthesis).
2. Run that method in long format only: apply the "> LONG FORMAT" section and its constraints (synthesis, three sections with subsections, financial-impact notations, governance-quality rating), plus the complementary synthesis and the sources. Deliver the Word report via scripts/render_esg_long.py following the Markdown contract in the "Long-format Word output" section, then present the .docx.
3. Do not run Step 0 (format choice) and do not ask the user which format they want: the format is fixed to long. Go straight to confirming the company name and the source(s), then to source collection.

Everything else (inputs, method, mandatory user check-ins, certifications glossary, complementary synthesis, sources, constraints, pitfalls) is defined in pe-esg-analysis and must be followed as-is.

Note: if you renamed the skills with a company prefix at install time (e.g. acme-pe-esg-analysis), use that same prefix in the path at step 1 (../acme-pe-esg-analysis/SKILL.md).

8. Long-format Word export module

The long format produces a styled Word document via a self-contained script. It builds the .docx from scratch - an automatic synthesis table, colour-coded title blocks with pillar pastilles (E / S / G) and coloured Exposure / Maturity cells - with **no external template and no logo**. Save the script below as `scripts/render_esg_long.py` in the main skill folder. The skill writes the analysis to a Markdown file following the contract in section 7.1, then runs:

```
python3 scripts/render_esg_long.py --content analysis.md --out "ESG Analysis - CompanyName.docx"
```

Requirements: Python 3 only (standard library, no third-party package). The output opens in Microsoft Word, LibreOffice and Google Docs. The colours are semantic (green / blue / grey / orange / red / purple), not brand colours; adjust the colour dictionaries at the top of the script to fit your own palette.

8.1 scripts/render_esg_long.py

```
# -*- coding: utf-8 -*-
"""Render a long ESG analysis to a self-contained Word (.docx) file.

No external template and no logo: the script builds the whole .docx from scratch,
including the synthesis table and the coloured title blocks.

Usage:
    python3 render_esg_long.py --content analysis.md --out "ESG Analysis - <Company>.docx"

The content file (analysis.md) follows the Markdown contract described in the
installation guide (section "Long-format Word export").
"""
import re, argparse, zipfile

def esc(t): return t.replace('&', '&').replace('<', '<').replace('>', '>')
BASE = '<w:rFonts w:ascii="Arial" w:hAnsi="Arial" w:cs="Arial"/>'
NAVY = '1F3864'

# Colour conventions (fill, text). Semantic, brand-neutral.
EXP0 = {'Strong': ('305496', 'FFFFFF'), 'Medium': ('5B9BD5', '000000'), 'Weak': ('F2F2F2', '000000'), '?': ('7030A0', 'FFFFFF')}
SOLU = {'Strong': ('70AD47', 'FFFFFF'), 'Medium': ('E2EFDA', '000000'), 'Weak': ('F2F2F2', '000000'), '?': ('7030A0', 'FFFFFF')}
MATU = {'Very advanced': ('70AD47', 'FFFFFF'), 'Advanced': ('E2EFDA', '000000'), 'Medium': ('FFF2CC', '000000'), 'Limited': ('ED7D31', 'FFFFFF'), 'Poor practices': ('C00000', 'FFFFFF'), '?': ('7030A0', 'FFFFFF')}
PILLAR = {'E': ('92D050', 'FFFFFF'), 'S': ('00B0F0', 'FFFFFF'), 'E/S': ('60D9E2', 'FFFFFF'), 'G': ('FF66FF', 'FFFFFF')}
ARROW_COL = {'++': ('70AD47', 'FFFFFF'), '+': ('E2EFDA', '000000'), '=': ('FFF2CC', '000000'), '--': ('ED7D31', 'FFFFFF'), '---': ('C00000', 'FFFFFF')}
COMBINE = {('Weak', 'Very advanced'): '++', ('Medium', 'Very advanced'): '++', ('Strong', 'Very advanced'): '+', ('Weak', 'Advanced'): '++', ('Medium', 'Advanced'): '+', ('Strong', 'Advanced'): '+', ('Weak', 'Medium'): '+', ('Medium', 'Medium'): '=', ('Strong', 'Medium'): '=', ('Weak', 'Limited'): '=', ('Medium', 'Limited'): '-', ('Strong', 'Limited'): '-', ('Weak', 'Poor practices'): '-', ('Medium', 'Poor practices'): '-', ('Strong', 'Poor practices'): '-'}
GOV_ARROW = {'Very advanced': '++', 'Advanced': '+', 'Medium': '=', 'Limited': '-', 'Poor practices': '--', '?': '='}
PROD_ARROW = {'Strong': '++', 'Medium': '+', 'Weak': '=', '?': '='}

def esc_run(text, rpr=''):
    return '<w:r><w:rPr>'+rpr+'</w:rPr><w:t xml:space="preserve">'+esc(text)+'</w:t></w:r>' if text else ''

def cellp(text, rpr='', jc=None, spacing=None):
    j = '<w:jc w:val="'+jc+'"/>' if jc else ''
    sp = spacing or ''
    return '<w:p><w:pPr><w:keepNext/><w:keepLines/>'+sp+j+'</w:pPr>'+esc_run(text, rpr)+'</w:p>'

def cell(w, fill, content_p, valign='center', gridspan=None):
    shd = '<w:shd w:val="clear" w:color="auto" w:fill="'+fill+'"/>' if fill else ''
    gs = '<w:gridSpan w:val="'+str(gridspan)+'"/>' if gridspan else ''
    return '<w:tc><w:tcPr><w:tcW w:w="'+str(w)+'" w:type="dxa"/>'+gs+shd+'<w:vAlign w:val="'+valign+'"/></w:tc>'

if __name__ == '__main__':
    parser = argparse.ArgumentParser()
    parser.add_argument('--content', required=True, help='Markdown file to render')
    parser.add_argument('--out', required=True, help='Output Word document file')
    args = parser.parse_args()

    with open(args.content, 'r') as f:
        content = f.read()

    # Render the document
    docx = render(content, args.out)
```

```

w:tblPr>'<content p+'</w:tc>
NOBORD = ('<w:tblBorders><w:top w:val="none" w:sz="0" w:space="0" w:color="auto"/><w:left w:val="none"
w:sz="0" w:space="0" w:color="auto"/><w:bottom w:val="none" w:sz="0" w:space="0" w:color="auto"/><w:right
w:val="none" w:sz="0" w:space="0" w:color="auto"/><w:insideH w:val="none" w:sz="0" w:space="0"
w:color="auto"/><w:insideV w:val="none" w:sz="0" w:space="0" w:color="auto"/></w:tblBorders>')
def tblpr(total, jc=None):
    j = '<w:jc w:val="'+jc+'"/>' if jc else ''
    return '<w:tblPr><w:tblW w:w="'+str(total)+'" w:type="dxa"/>'+j+NOBORD+'<w:tblLook w:val="04A0"
w:firstRow="1" w:lastRow="0" w:firstColumn="1" w:lastColumn="0" w:noHBand="0" w:noVBand="1"/></w:tblPr>'
HDR_RPR = BASE+'<w:b/><w:bCs/><w:sz w:val="16"/><w:szCs w:val="16"/>'
HDR_SP = '<w:spacing w:before="0" w:after="40"/>'
EMPTY_SP= '<w:spacing w:before="0" w:after="0"/>'
ROWH = '<w:trPr><w:cantSplit/><w:trHeight w:val="397" w:hRule="atLeast"/></w:trPr>'
HROW = '<w:tr><w:trPr><w:cantSplit/></w:trPr>'

def body_para(text):
    return '<w:p><w:pPr><w:jc w:val="both"/><w:spacing w:after="120"/></w:pPr>'+esc_run(text, BASE)+'</
w:p>'
def kicker(text):
    return '<w:p><w:pPr><w:spacing w:after="0"/></w:pPr><w:r><w:rPr>'+BASE+'<w:caps/><w:color
w:val="'+NAVY+'"/><w:sz w:val="16"/><w:szCs w:val="16"/></w:rPr><w:t xml:space="preserve">'+esc(text)+'</
w:t></w:r></w:p>'
def title_para(text):
    return '<w:p><w:pPr><w:spacing w:before="40" w:after="40"/></w:pPr><w:r><w:rPr>'+BASE+'<w:b/><w:bCs/
><w:color w:val="0B3142"/><w:sz w:val="38"/><w:szCs w:val="38"/></w:rPr><w:t
xml:space="preserve">'+esc(text)+'</w:t></w:r></w:p>'
def subtitle_para(text):
    return '<w:p><w:pPr><w:spacing w:after="240"/></w:pPr><w:r><w:rPr>'+BASE+'<w:i/><w:iCs/><w:color
w:val="5A5A5A"/></w:rPr><w:t xml:space="preserve">'+esc(text)+'</w:t></w:r></w:p>'
def h1_para(text, pagebreak=False):
    pb = '<w:pageBreakBefore/>' if pagebreak else ''
    bdr = '<w:pBdr><w:bottom w:val="single" w:sz="8" w:space="2" w:color="'+NAVY+'"/></w:pBdr>'
    return '<w:p><w:pPr>'+pb+'<w:keepNext/>'+bdr+'<w:spacing w:before="260" w:after="100"/></
w:pPr><w:r><w:rPr>'+BASE+'<w:b/><w:bCs/><w:color w:val="'+NAVY+'"/><w:sz w:val="27"/><w:szCs w:val="27"/
></w:rPr><w:t xml:space="preserve">'+esc(text)+'</w:t></w:r></w:p>'
def h2_para(text):
    return '<w:p><w:pPr><w:keepNext/><w:spacing w:before="160" w:after="40"/></
w:pPr><w:r><w:rPr>'+BASE+'<w:b/><w:bCs/><w:color w:val="2B4A7E"/><w:sz w:val="22"/><w:szCs w:val="22"/></
w:rPr><w:t xml:space="preserve">'+esc(text)+'</w:t></w:r></w:p>'
def impact_line(text):
    return ('<w:p><w:pPr><w:keepNext/><w:keepLines/><w:spacing w:before="360" w:after="60"/></
w:pPr><w:r><w:rPr><w:i/><w:iCs/>'+BASE+'</w:rPr><w:tab/><w:t xml:space="preserve">'+esc(text)+'</w:t></
w:r></w:p>')

def risk_table(letter, title, expo, matu):
    pc,pt = PILLAR.get(letter,PILLAR['S']); ec,et = EXPO.get(expo,EXPO['?']); mc,mt =
MATU.get(matu,MATU['?'])
    W1,W2,W3,W4 = 620,5762,1380,1308
    grid = '<w:tblGrid><w:gridCol w:w="%d"/><w:gridCol w:w="%d"/><w:gridCol w:w="%d"/><w:gridCol w:w="%d"/
></w:tblGrid>'%(W1,W2,W3,W4)
    row1 = (HROW+cell(W1,'',cellp('', BASE+'<w:sz w:val="16"/><w:szCs w:val="16"/>', spacing=EMPTY_SP))
    + cell(W2,'',cellp('', BASE+'<w:sz w:val="16"/><w:szCs w:val="16"/>', spacing=EMPTY_SP))
    + cell(W3,'',cellp('Exposure', HDR_RPR,'center',spacing=HDR_SP))
    + cell(W4,'',cellp('Maturity', HDR_RPR,'center',spacing=HDR_SP))+'</w:tr>')
    row2 = ('<w:tr>'+ROWH
    + cell(W1,pc,cellp(letter, BASE+'<w:b/><w:bCs/><w:color w:val="'+pt+'"/
>','center',spacing=EMPTY_SP))
    + cell(W2,'D9D9D9',cellp(title, BASE+'<w:b/><w:bCs/>', 'left',spacing=EMPTY_SP+'<w:ind
w:left="170"/>'))
    + cell(W3,ec,cellp(expo, BASE+'<w:color w:val="'+et+'"/><w:sz w:val="16"/><w:szCs w:val="16"/
>','center',spacing=EMPTY_SP))
    + cell(W4,mc,cellp(matu, BASE+'<w:color w:val="'+mt+'"/><w:sz w:val="16"/><w:szCs w:val="16"/
>','center',spacing=EMPTY_SP))+'</w:tr>')
    return '<w:tbl>'+tblpr(W1+W2+W3+W4)+grid+row1+row2+'</w:tbl>'

def gov_table(title, matu):
    pc,pt = PILLAR['G']; mc,mt = MATU.get(matu,MATU['?'])
    W1,W2,W3 = 620,7142,1308
    grid = '<w:tblGrid><w:gridCol w:w="%d"/><w:gridCol w:w="%d"/><w:gridCol w:w="%d"/></w:tblGrid>'%(
W1,W2,W3)
    row1 = (HROW+cell(W1,'',cellp('', BASE+'<w:sz w:val="16"/><w:szCs w:val="16"/>', spacing=EMPTY_SP))
    + cell(W2,'',cellp('', BASE+'<w:sz w:val="16"/><w:szCs w:val="16"/>', spacing=EMPTY_SP))
    + cell(W3,'',cellp('Maturity', HDR_RPR,'center',spacing=HDR_SP))+'</w:tr>')
    row2 = ('<w:tr>'+ROWH
    + cell(W1,pc,cellp('G', BASE+'<w:b/><w:bCs/><w:color w:val="'+pt+'"/>', 'center',spacing=EMPTY_SP))
    + cell(W2,'D9D9D9',cellp(title, BASE+'<w:b/><w:bCs/>', 'left',spacing=EMPTY_SP+'<w:ind

```



```

w:left="170"/>')
    + cell(W3,mc,cellp(matu, BASE+'<w:color w:val="'+mt+'"/><w:sz w:val="16"/><w:szCs w:val="16"/
>','center',spacing=EMPTY_SP))+'</w:tr>')
    return '<w:tbl>'+tblpr(W1+W2+W3)+grid+row1+row2+'</w:tbl>'

def products_table(statement, rating, pillar, expo_type, themes):
    rc,rt = SOLU.get(rating,SOLU['?']); pc,pt = PILLAR.get(pillar,PILLAR['E'])
    PB = BASE+'<w:sz w:val="18"/><w:szCs w:val="18"/>'
    C1,C2,C3,C4,C5,C6,C7 = 3827,150,900,120,520,120,880
    grid = '<w:tblGrid>'+''.join('<w:gridCol w:w="%d"/>%w for w in [C1,C2,C3,C4,C5,C6,C7])+'</w:tblGrid>'
    def vc(w, content='', fill='', vmerge=None, gridspan=None):
        vm = '<w:vMerge w:val="restart"/>' if vmerge=='restart' else ('<w:vMerge/>' if vmerge=='cont' else
        '')
        shd = '<w:shd w:val="clear" w:color="auto" w:fill="'+fill+'"/>' if fill else ''
        gs = '<w:gridSpan w:val="'+str(gridspan)+'"/>' if gridspan else ''
        return '<w:tc><w:tcPr><w:tcW w:w="'+str(w)+'" w:type="dxa"/>'+gs+vm+shd+'<w:vAlign w:val="center"/
></w:tcPr>'+(content or '<w:p><w:pPr><w:keepNext/><w:keepLines/><w:pPr><w:p>')+'</w:tc>'
    P = lambda txt,rpr=PB,jc=None: cellp(txt, rpr, jc, spacing=EMPTY_SP)
    R1 = '<w:trPr><w:cantSplit/><w:trHeight w:val="150" w:hRule="atLeast"/></w:trPr>'
    RC = '<w:trPr><w:cantSplit/><w:trHeight w:val="260" w:hRule="atLeast"/></w:trPr>'
    r1 = ('<w:tr>'+R1+vc(C1, P(statement,PB,'left'), vmerge='restart')+vc(C2, P(':',PB,'center'),
vmerge='restart')
    + vc(C3)+vc(C4)+vc(C5)+vc(C6)+vc(C7, P(expo_type,PB,'center'), vmerge='restart')+'</w:tr>')
    r2 = ('<w:tr>'+RC+vc(C1, vmerge='cont')+vc(C2, vmerge='cont')
    + vc(C3, P(rating, PB+'<w:color w:val="'+rt+'"/>','center'), fill=rc)+vc(C4)
    + vc(C5, P(pillar, PB+'<w:b/><w:bCs/><w:color w:val="'+pt+'"/>','center'), fill=pc)+vc(C6)
    + vc(C7, vmerge='cont')+'</w:tr>')
    r3 = ('<w:tr>'+R1+vc(C1, vmerge='cont')+vc(C2, vmerge='cont')+vc(C3)+vc(C4)+vc(C5)+vc(C6)+vc(C7,
vmerge='cont')+'</w:tr>')
    rtth = ('<w:tr><w:trPr><w:cantSplit/></w:trPr>'+vc(C1, P('Theme(s)',PB,'left'))+vc(C2,
P(':',PB,'center'))
    + vc(C3+C4+C5+C6+C7, P(themes,PB,'left'), gridspan=5)+'</w:tr>')
    return '<w:tbl>'+tblpr(C1+C2+C3+C4+C5+C6+C7,'center')+grid+r1+r2+r3+rtth+'</w:tbl>'

S1,S2,S3,S4,S5,S6 = 720,5471,80,620,120,360
STOT = S1+S2+S3+S4+S5+S6
SROWH = '<w:trPr><w:cantSplit/><w:trHeight w:val="312" w:hRule="atLeast"/></w:trPr>'
SEP = '<w:tr><w:trPr><w:trHeight w:val="80" w:hRule="exact"/></w:trPr>'+cell(STOT,'',cellp('', BASE+'<w:sz
w:val="2"/><w:szCs w:val="2"/>', spacing=EMPTY_SP), gridspan=6)+'</w:tr>'
def synth_row(letter, title, arrow, dollar):
    pc,pt = PILLAR.get(letter,PILLAR['S']); ac,at = ARROW_COL.get(arrow,ARROW_COL['='])
    if dollar=='pos': dol = cell(S6,'70AD47',cellp('$', BASE+'<w:b/><w:bCs/><w:color w:val="FFFFFF"/
>','center',spacing=EMPTY_SP))
    elif dollar=='neg': dol = cell(S6,'ED7D31',cellp('$', BASE+'<w:b/><w:bCs/><w:color w:val="FFFFFF"/
>','center',spacing=EMPTY_SP))
    else: dol = cell(S6,'',cellp('',BASE,spacing=EMPTY_SP))
    return ('<w:tr>'+SROWH
    + cell(S1,pc,cellp(letter, BASE+'<w:b/><w:bCs/><w:color w:val="'+pt+'"/
>','center',spacing=EMPTY_SP))
    + cell(S2,'',cellp(title, BASE,'left',spacing=EMPTY_SP))
    + cell(S3,'',cellp('',BASE,spacing=EMPTY_SP))
    + cell(S4,ac,cellp(arrow, BASE+'<w:b/><w:bCs/><w:color w:val="'+at+'"/
>','center',spacing=EMPTY_SP))
    + cell(S5,'',cellp('',BASE,spacing=EMPTY_SP))+dol+'</w:tr>')
def synth_sub(text):
    return ('<w:tr><w:trPr><w:cantSplit/><w:trHeight w:val="200" w:hRule="atLeast"/></w:trPr>'
    + cell(STOT,'',cellp(text, BASE+'<w:i/><w:iCs/>', 'left',spacing='<w:spacing w:before="240"
w:after="40"/>'), gridspan=6)+'</w:tr>')
def chip(sym, fill, txt, tail=''):
    return '<w:r><w:rPr>'+BASE+'<w:b/><w:bCs/><w:color w:val="'+txt+'"/><w:shd w:val="clear"
w:color="auto" w:fill="'+fill+'"/></w:rPr><w:t xml:space="preserve"> '+esc(sym)+' '+tail+'</w:t></w:r>'
def gap(): return '<w:r><w:rPr>'+BASE+'</w:rPr><w:t xml:space="preserve"> </w:t></w:r>'
def lbl(t): return '<w:r><w:rPr>'+BASE+'</w:rPr><w:t xml:space="preserve">'+esc(t)+'</w:t></w:r>'
def lbl2(t): return '<w:r><w:rPr>'+BASE+'<w:i/><w:iCs/><w:sz w:val="18"/><w:szCs w:val="18"/></w:rPr><w:t
xml:space="preserve">'+esc(t)+'</w:t></w:r>'
ANCHOR = '<w:r><w:rPr>'+BASE+'</w:rPr><w:t xml:space="preserve">'+chr(0x200b)+'</w:t></w:r>'
def synth_table(product, risks, govs):
    rows = []
    if product:
        plabel = product.get('label','Impacts of products and services')
        if product.get('main_theme'): plabel += ': ' + product['main_theme']
        rows += [synth_row(product['pillar'], plabel, product['arrow'], product['dollar'])]
    if risks:
        rows += [SEP, synth_sub('Main environmental and social risks')]
        for r in risks: rows += [SEP, synth_row(r['pillar'], r['title'], r['arrow'], r['dollar'])]
    if govs:

```

```

        rows += [SEP, synth_sub('Governance'')]
        for g in govs: rows += [SEP, synth_row('G', g['title'], g['arrow'], g['dollar'])]
    grid = '<w:tblGrid>'+''.join('<w:gridCol w:w="%d"/>' % w for w in [S1,S2,S3,S4,S5,S6])+'</w:tblGrid>'
    tbl = '<w:tbl>' +tblpr(STOT, 'center') +grid+''.join(rows)+'</w:tbl>'
    leg1 = ('<w:p><w:pPr><w:spacing w:before="0" w:after="80"/><w:jc w:val="center"/></w:pPr>' +lbl('ESG
Maturity scale: '))
    + chip('++', '70AD47', 'FFFFFF') +gap() +chip('+', 'E2EFDA', '000000') +gap() +chip('=', 'FFF2CC', '000000')
+gap()
    + chip('-', 'ED7D31', 'FFFFFF') +gap() +chip('--', 'C00000', 'FFFFFF') +ANCHOR+'</w:p>')
    leg2 = ('<w:p><w:pPr><w:spacing w:before="80" w:after="120"/><w:jc w:val="center"/></w:pPr>'
    + lbl2('Potential positive financial impacts: ') +chip('$', '70AD47', 'FFFFFF') +gap() +gap()
    + lbl2('Potential negative financial impacts: ') +chip('$', 'ED7D31', 'FFFFFF') +ANCHOR+'</w:p>')
    return '<w:p>' +tbl+'<w:p/>' +leg1+leg2+'<w:p/>'

def md_table(raw_rows):
    rows = []
    for r in raw_rows:
        cells = [c.strip() for c in r.strip().strip('|').split('|')]
        if cells and all(re.fullmatch(r'?:-{2,}:?', (c or '-')) for c in cells):
            continue
        rows.append(cells)
    if not rows: return ''
    ncols = max(len(r) for r in rows)
    total = 9360; colw = total // ncols
    grid = '<w:tblGrid>'+''.join('<w:gridCol w:w="%d"/>' % colw for _ in range(ncols))+'</w:tblGrid>'
    bord = ('<w:tblBorders>'+''.join('<w:s w:val="single" w:sz="4" w:space="0" w:color="C8CDD6"/>' % e for
e in ['top', 'left', 'bottom', 'right', 'insideH', 'insideV']))+'</w:tblBorders>')
    tpr = '<w:tblPr><w:tblW w:w="%d" w:type="dxa"/>' % s<w:tblLook w:val="04A0"/></w:tblPr>' % (total, bord)
    out_rows = ''
    for i, r in enumerate(rows):
        cells = r + [''] * (ncols - len(r))
        hdr = (i==0)
        rpr = BASE+'<w:sz w:val="17"/><w:szCs w:val="17"/>' + ('<w:b/><w:bCs/><w:color w:val="FFFFFF"/>' if
hdr else '')
        fill = '1F3864' if hdr else ''
        tr = '<w:tr><w:trPr><w:cantSplit/></w:trPr>'
        for c in cells:
            tr += cell(colw, fill, cellp(c, rpr, 'left', spacing='<w:spacing w:before="10" w:after="10"/
>'), valign='center')
            tr += '</w:tr>'
        out_rows += tr
    return '<w:p>' +tbl+'<w:p/>' +tpr+grid+out_rows+'</w:tbl><w:p/>'

def find_override(t):
    o = re.search(r'Synthesis\s*:\s*(\+|\+|--|\+|=|-)', t)
    return o.group(1) if o else None
def parse_risk_notation(t):
    m = re.search(r'Exposure\s*:\s*([^\]]+?)\s*\| \s*Maturity\s*:\s*([^\]]+?)\s*([^\]]+?)', t)
    expo = m.group(1).strip() if m else '?'; matu = m.group(2).strip() if m else '?'
    clean = re.sub(r'\s*(Exposure.*?)\s*$', '', t).strip()
    return clean, expo, matu, find_override(t)
def parse_matu(t):
    m = re.search(r'Maturity\s*:\s*([^\]]+?)\s*([^\]]+?)', t)
    matu = m.group(1).strip() if m else '?'
    clean = re.sub(r'\s*(Maturity.*?)\s*$', '', t).strip()
    return clean, matu, find_override(t)
PILL_RE = re.compile(r'^\[(E|S|E|S|G)\]\s*')
def sign_to_dollar(t):
    m = re.search(r',\s*([+|=])\s*$', t)
    if not m: return None
    return {'+': 'pos', '-': 'neg', '=': None}.get(m.group(1))

def parse(content):
    lines = content.split('\n')
    title = subtitle = None
    elements = []
    product = None; risks = []; govs = []; current = None
    buf = []
    in_table = False; tbl_rows = []
    def flush():
        nonlocal buf
        txt = ' '.join(x.strip() for x in buf if x.strip())
        if txt: elements.append(('para', txt))
        buf = []
    for ln in lines:
        s = ln.strip()

```

```

if s.startswith('|'):
    flush(); tbl_rows.append(s); in_table = True; continue
if in_table:
    elements.append(('table', tbl_rows)); tbl_rows = []; in_table = False
mt = re.match(r'^TITLE\s*:\s*(.+)$', s)
ms = re.match(r'^SUBTITLE\s*:\s*(.+)$', s)
me = re.match(r'^\[HEADER\]\s*(.+)$', s, re.I)
if mt: title = mt.group(1).strip(); continue
if ms: subtitle = ms.group(1).strip(); continue
if me:
    flush()
    kv = {}
    for part in me.group(1).split('|'):
        if '=' in part:
            k,v = part.split('=',1); kv[k.strip().lower()] = v.strip()
            note = kv.get('note','?'); pil = kv.get('pillar','E'); ex = kv.get('exposure','')
            th = kv.get('themes','')
            ovr = kv.get('synthesis')
            scope = (kv.get('scope','both') or 'both').strip().lower()
            scope_label = {'products':'Impacts of products','services':'Impacts of
services'}.get(scope,'Impacts of products and services')
            scope_word = {'products':'products','services':'services'}.get(scope,'products and services')
            main_theme = th.split(',')[0].strip() if th else ''
            product = {'pillar':pil,'arrow':ovr or PROD_ARROW.get(note,''),'dollar':None,
                        'statement':kv.get('statement','Exposure to '+scope_word+' with a positive impact
on environmental and social themes'),
                        'note':note,'expo_type':ex,'themes':th,'label':scope_label,'main_theme':main_theme}
            current = product
            elements.append(('products', product)); continue
if s.startswith('### ') or re.match(r'^#{2,3}\s+Potential financial impacts', s, re.I):
    inner = re.sub(r'^#{1,6}\s+', '', s)
    if inner.lower().startswith('potential financial impacts'):
        flush()
        d = sign_to_dollar(inner)
        if current is not None and current.get('dollar') is None and d: current['dollar'] = d
        elements.append(('impact', inner)); continue
if s.startswith('## '):
    flush()
    inner = s[3:].strip()
    mp = PILL_RE.match(inner)
    if mp and mp.group(1) == 'G':
        rest = PILL_RE.sub('', inner)
        if re.search(r'Exposure\s*:', rest):
            ttl, expo, matu, ovr = parse_risk_notation(rest)
            e = {'title':ttl,'arrow':ovr or
COMBINE.get((expo,matu),''),'dollar':None,'matu':matu,'expo':expo}
        else:
            ttl, matu, ovr = parse_matu(rest)
            e = {'title':ttl,'arrow':ovr or GOV_ARROW.get(matu,''),'dollar':None,'matu':matu}
            govs.append(e); current = e
            elements.append(('gov', e)); continue
    if mp:
        pil = mp.group(1); rest = PILL_RE.sub('', inner)
        ttl, expo, matu, ovr = parse_risk_notation(rest)
        e = {'pillar':pil,'title':ttl,'arrow':ovr or
COMBINE.get((expo,matu),''),'dollar':None,'expo':expo,'matu':matu}
        risks.append(e); current = e
        elements.append(('risk', e)); continue
    elements.append(('h2', inner)); continue
if s.startswith('# '):
    flush()
    inner = s[2:].strip()
    inner = re.sub(r'\s*(Governance quality.*?)\s*$', '', inner).strip()
    elements.append(('h1', inner)); continue
if s == '':
    flush(); continue
buf.append(ln)
if in_table: elements.append(('table', tbl_rows))
flush()
return title, subtitle, elements, product, risks, govs

def render_body(content):
    title, subtitle, elements, product, risks, govs = parse(content)
    if not title: raise SystemExit("ERROR: missing 'TITLE: <Company>' line in the content.")
    out = []
    out.append(kicker('ESG Analysis'))

```

```

out.append(title_para(title))
if subtitle: out.append(subtitle_para(subtitle))
after_block = False
for el in elements:
    k = el[0]
    if k == 'h1':
        t = el[1]
        pb = t.lower().startswith('complementary synthesis')
        out.append(h1_para(t, pagebreak=pb))
        if t.lower() == 'synthesis':
            out.append(synth_table(product, risks, gov))
            after_block = False
    elif k == 'h2':
        out.append(h2_para(el[1])); after_block = False
    elif k == 'products':
        p = el[1]; out.append(products_table(p['statement'], p['note'], p['pillar'], p['expo_type'],
p['themes']))
        out.append('<w:p/>'); after_block = False
    elif k == 'risk':
        e = el[1]; out.append(risk_table(e['pillar'], e['title'], e['expo'], e['matu'])); after_block
= True
    elif k == 'gov':
        e = el[1]
        if e.get('expo'):
            out.append(risk_table('G', e['title'], e['expo'], e['matu']))
        else:
            out.append(gov_table(e['title'], e['matu']))
            after_block = True
    elif k == 'impact':
        out.append(impact_line(el[1])); after_block = False
    elif k == 'table':
        out.append(md_table(el[1])); after_block = False
    elif k == 'para':
        out.append(body_para(el[1]) if not after_block else body_para_sb(el[1])); after_block = False
return '\n'.join(out)

def body_para_sb(text):
    return '<w:p><w:pPr><w:jc w:val="both"/><w:spacing w:before="120" w:after="120"/></
w:pPr>'+esc_run(text, BASE)+'</w:p>'

WNS = ('xmlns:w="http://schemas.openxmlformats.org/wordprocessingml/2006/main" '
'xmlns:r="http://schemas.openxmlformats.org/officeDocument/2006/relationships"')
SECTPR = '<w:sectPr><w:pgSz w:w="11906" w:h="16838"/><w:pgMar w:top="1134" w:right="1134" w:bottom="1134"
w:left="1134" w:header="708" w:footer="708" w:gutter="0"/></w:sectPr>'
CONTENT_TYPES = ('<?xml version="1.0" encoding="UTF-8" standalone="yes"?>'
'<Types xmlns="http://schemas.openxmlformats.org/package/2006/content-types">'
'<Default Extension="rels" ContentType="application/vnd.openxmlformats-package.relationships+xml"/>'
'<Default Extension="xml" ContentType="application/xml"/>'
'<Override PartName="/word/document.xml" ContentType="application/vnd.openxmlformats-
officedocument.wordprocessingml.document.main+xml"/>'
'<Override PartName="/word/styles.xml" ContentType="application/vnd.openxmlformats-
officedocument.wordprocessingml.styles+xml"/>'
'</Types>')
RELS = ('<?xml version="1.0" encoding="UTF-8" standalone="yes"?>'
'<Relationships xmlns="http://schemas.openxmlformats.org/package/2006/relationships">'
'<Relationship Id="rId1" Type="http://schemas.openxmlformats.org/officeDocument/2006/relationships/
officeDocument" Target="word/document.xml"/>'
'</Relationships>')
DOC_RELS = ('<?xml version="1.0" encoding="UTF-8" standalone="yes"?>'
'<Relationships xmlns="http://schemas.openxmlformats.org/package/2006/relationships">'
'<Relationship Id="rId1" Type="http://schemas.openxmlformats.org/officeDocument/2006/relationships/
styles" Target="styles.xml"/>'
'</Relationships>')
STYLES = ('<?xml version="1.0" encoding="UTF-8" standalone="yes"?>'
'<w:styles '+WNS+'><w:docDefaults><w:rPrDefault><w:rPr>'+BASE+'<w:sz w:val="20"/><w:szCs w:val="20"/></
w:rPr></w:rPrDefault></w:docDefaults>'
'<w:style w:type="paragraph" w:default="1" w:styleId="Normal"><w:name w:val="Normal"/></w:style></
w:styles>')

def build_docx(out_path, body_xml):
    document = ('<?xml version="1.0" encoding="UTF-8" standalone="yes"?>'
'<w:document '+WNS+'><w:body>'+body_xml+SECTPR+'</w:body></w:document>')
    with zipfile.ZipFile(out_path, 'w', zipfile.ZIP_DEFLATED) as z:
        z.writestr('[Content_Types].xml', CONTENT_TYPES)
        z.writestr('_rels/.rels', RELS)
        z.writestr('word/_rels/document.xml.rels', DOC_RELS)

```

```
z.writestr('word/styles.xml', STYLES)
z.writestr('word/document.xml', document)

def main():
    ap = argparse.ArgumentParser()
    ap.add_argument('--content', required=True)
    ap.add_argument('--out', required=True)
    a = ap.parse_args()
    content = open(a.content, encoding='utf-8').read()
    body_xml = render_body(content)
    build_docx(a.out, body_xml)
    print('OK ->', a.out)

if __name__ == '__main__':
    main()
```

Appendix A - Product and service impact themes

Save this list as `references/impact-themes.md` in the main skill folder. When describing the impacts of a company's products or services, the skill picks the impact theme(s) from this list and uses **only the theme names** in the analysis. The sub-theme examples are there **solely to guide theme selection** and must never be reproduced in the output. In the synthesis table, the products/services row shows the section name followed by the main theme (e.g. "Impacts of services: Education & knowledge").

Low-carbon energy (examples, for selection guidance only, not to be reproduced: Solar, wind, hydro, biomass, geothermal, smart grids)

Low-carbon mobility (examples, for selection guidance only, not to be reproduced: Electric / hydrogen vehicles, public transportation, cycles, route optimization)

Green buildings (examples, for selection guidance only, not to be reproduced: Insulation, low-carbon heating and air conditioning, intelligent consumption management, green building materials)

Green industry (examples, for selection guidance only, not to be reproduced: Energy efficiency of industrial processes, green chemistry, circularity)

Sustainable food (examples, for selection guidance only, not to be reproduced: Certified agriculture (organic, fair trade), precision agriculture, plant-based food)

Nature-based solutions (examples, for selection guidance only, not to be reproduced: Reforestation, forest protection, vegetation)

Sustainable waste & water management (examples, for selection guidance only, not to be reproduced: Waste sorting & recycling, leak reduction, smart irrigation, access to clean water, sanitation and waste management)

Sustainable consumption (examples, for selection guidance only, not to be reproduced: Certified products (organic, fair trade), eco-design, circular economy, packaging reduction, alternatives to plastic, sustainable raw materials / ingredients, eco-tourism)

Better health for all (examples, for selection guidance only, not to be reproduced: Diagnostics, treatments, prevention, care, medical equipment, access to health)

Education & knowledge (examples, for selection guidance only, not to be reproduced: Education, lifelong learning, knowledge bases, access to education & information, courses on sustainability)

Social housing (examples, for selection guidance only, not to be reproduced: Development, maintenance & dedicated services)

Safety (examples, for selection guidance only, not to be reproduced: Dedicated safety equipment (e.g. airbags, belts, protective masks/suits), testing (e.g. health, food, industry), cyber security)

Tackling inequalities (examples, for selection guidance only, not to be reproduced: Solutions in favour of gender diversity (e.g. childcare solutions, contraception), people with disabilities, access to quality employment, access to essential services)

Sustainable ICT (examples, for selection guidance only, not to be reproduced: Energy-efficient hardware, energy-efficient data centres, ESG and EQSH software, dematerialisation, new technologies fostering sustainability)

Sustainable finance (examples, for selection guidance only, not to be reproduced: Micro-credit, sustainable finance, access to financial services)

Appendix B - SASB sector reference

Save this list as `references/sasb-disclosure-topics.md` in the main skill folder. It is the simplified SASB materiality list: 77 industries across 11 macro-sectors, each with its disclosure topics. Source: SASB Standards Navigator (Materiality Finder), IFRS Foundation - public data. The skill reads only the target's sector block. A one-line description is shown only for the less self-explanatory topics; for the others the topic name is enough. Topic labels are in English (official SASB terminology); transpose them into your analysis language.

Consumer Goods

1. Apparel, Accessories & Footwear

- Management of Chemicals in Products
- Environmental Impacts in the Supply Chain
- Labour Conditions in the Supply Chain
- Raw Materials Sourcing

2. Appliance Manufacturing

- Product Safety
- Product Lifecycle Environmental Impacts** : Entities in the Appliance Manufacturing industry seek to differentiate their products from those of competitors.

3. Building Products & Furnishings

- Energy Management in Manufacturing
- Management of Chemicals in Products
- Product Lifecycle Environmental Impacts** : Depending on the specific building product or furnishing, significant environmental impacts can arise during raw material sourcing, transportation, manufacturing, use-phase or end-of-life.
- Wood Supply Chain Management

4. E-Commerce

- Hardware Infrastructure Energy & Water Management
- Data Privacy & Advertising Standards
- Data Security
- Employee Recruitment, Inclusion & Performance
- Product Packaging & Distribution** : A significant part of the E-Commerce industry's added value comes from an entity's ability to move a wide array of goods efficiently to consumers who would otherwise have to personally travel to collect the...

5. Household & Personal Products

- Water Management
- Product Environmental, Health and Safety Performance
- Packaging Lifecycle Management** : The Household & Personal Products industry uses a large quantity of materials for product packaging, which often constitutes a significant portion of entities' expenses.
- Environmental & Social Impacts of Palm Oil Supply Chain

6. Multiline and Specialty Retailers & Distributors

- Energy Management in Retail & Distribution
- Data Security
- Labour Practices
- Workforce Diversity & Inclusion
- Product Sourcing, Packaging & Marketing** : Entities in the Multiline and Specialty Retailers & Distributors industry sell a wide array of products including electronics, clothing, furnishings and cosmetics, all of which have environmental and social...

7. Toys & Sporting Goods

- Chemical & Safety Hazards of Products
- Labour Conditions in the Supply Chain

Extractives & Minerals Processing

8. Coal Operations

Greenhouse Gas Emissions

Water Management

Waste Management

Biodiversity Impacts

Rights of Indigenous Peoples

Community Relations

Labour Relations

Workforce Health & Safety

Reserves Valuation & Capital Expenditures : Coal entities may be unable to extract a significant proportion of their coal reserves if greenhouse gas (GHG) emissions are controlled to limit global temperature increases.

Tailings Storage Facilities Management : Coal waste impoundments or fine coal refuse ponds, also called tailings storage facilities (TSFs), can leak and contaminate water supplies if mismanaged, potentially leading to adverse impacts on the...

9. Construction Materials

Greenhouse Gas Emissions

Air Quality

Energy Management

Water Management

Waste Management

Biodiversity Impacts

Workforce Health & Safety

Product Innovation : Innovations in building materials are an essential component in the growth of sustainable construction.

Pricing Integrity & Transparency

10. Iron & Steel Producers

Greenhouse Gas Emissions

Air Quality

Energy Management

Water Management

Waste Management

Workforce Health & Safety

Supply Chain Management

11. Metals & Mining

Greenhouse Gas Emissions

Air Quality

Energy Management

Water Management

Waste & Hazardous Materials Management

Biodiversity Impacts

Security, Human Rights & Rights of Indigenous Peoples

Community Relations

Labour Practices

Workforce Health & Safety

Business Ethics & Transparency

Tailings Storage Facilities Management : The Metals & Mining industry faces significant operational hazards, particularly those associated with the structural integrity of tailings storage facilities (TSFs).

12. Oil & Gas - Exploration & Production

Greenhouse Gas Emissions

Air Quality

Water Management

Biodiversity Impacts

Security, Human Rights & Rights of Indigenous Peoples

Community Relations

Workforce Health & Safety

Reserves Valuation & Capital Expenditures : Exploration and production (E&P) entities may be unable to extract a significant proportion of their proved and probable oil and gas reserves if greenhouse gas (GHG) emissions are controlled to limit global...

Business Ethics & Transparency

Management of the Legal & Regulatory Environment : The Exploration & Production (E&P) industry is subject to numerous sustainability-related regulations and a rapidly changing regulatory environment.

Critical Incident Risk Management : The Exploration & Production (E&P) industry faces significant hazards associated with exploration, development and production activities.

13. Oil & Gas – Midstream

Greenhouse Gas Emissions

Air Quality

Ecological Impacts

Competitive Behaviour

Operational Safety, Emergency Preparedness & Response : Entities in the Oil & Gas – Midstream industry operate a vast network of assets at risk of spills and accidents.

14. Oil & Gas – Refining & Marketing

Greenhouse Gas Emissions

Air Quality

Water Management

Hazardous Materials Management

Workforce Health & Safety

Product Specifications & Clean Fuel Blends : Some regulatory jurisdictions have implemented product specifications and renewable fuel blends, which pose significant compliance and operational risks for Refining & Marketing (R&M) entities.

Pricing Integrity & Transparency

Management of the Legal & Regulatory Environment : The Refining & Marketing (R&M) industry is subject to numerous sustainability-related regulations and an often rapidly changing regulatory environment.

Critical Incident Risk Management : The operations of Refining & Marketing (R&M) entities are often characterised by a high number of hazards, including the handling of flammable, volatile substances, the use of highly reactive chemicals, and...

15. Oil & Gas – Services

Emissions Reduction Services & Fuels Management

Water Management Services

Chemicals Management

Ecological Impact Management

Workforce Health & Safety

Business Ethics & Payments Transparency

Management of the Legal & Regulatory Environment : The Oil & Gas – Services industry is subject to numerous sustainability-related regulations and a rapidly changing regulatory environment.

Critical Incident Risk Management : Entities in the Oil & Gas – Services industry are subject to significant risks associated with low-probability, high-consequence events associated with oil and gas exploration, development and production...

Financials

16. Asset Management & Custody Activities

Transparent Information & Fair Advice for Customers : Asset managers have legal obligations and fiduciary duties related to record keeping, operating and marketing, disclosure requirements and prevention of fraudulent activities.

Employee Diversity & Inclusion

Incorporation of Environmental, Social, and Governance Factors in Investment Management & Advisory : Asset Management & Custody Activities entities maintain a fiduciary responsibility to their clients.

Financed Emissions : Entities participating in asset management activities face risks and opportunities related to the greenhouse gas emissions associated with those activities.

Business Ethics

17. Commercial Banks

Data Security

Financial Inclusion & Capacity Building

Incorporation of Environmental, Social, and Governance Factors in Credit Analysis : As financial intermediaries, commercial banks contribute to significant positive and negative environmental and social externalities through their lending practices.

Financed Emissions : Entities participating in commercial banking activities face risks and opportunities related to the greenhouse gas emissions associated with those activities.

Business Ethics

Systemic Risk Management : Commercial Bank entities that fail to manage risks to capital effectively may suffer significant losses while increasing their liabilities.

18. Consumer Finance

Customer Privacy

Data Security

Selling Practices : Selling practices encompasses performance in three important areas that can affect an entity's operations and financial condition.

19. Insurance

Transparent Information & Fair Advice for Customers : Insurance products play an important societal role in alleviating unexpected economic shocks, allowing individual policyholders to reduce the financial consequences of events such as illnesses, accidents and...

Incorporation of Environmental, Social and Governance Factors in Investment Management : Insurance entities must invest capital to preserve accumulated premium revenues equivalent to expected policy claim pay-outs and maintain long-term asset-liability parity.

Policies Designed to Incentivise Responsible Behaviour : Advances in technology and the development of new policy products have allowed insurance entities to limit claim payments while encouraging responsible behaviour.

Financed Emissions : Entities participating in insurance activities face risks and opportunities related to the greenhouse gas emissions associated with those activities.

Physical Risk Exposure

Systemic Risk Management : Entities in the Insurance industry have the potential to pose, amplify or transmit a threat to the financial system.

20. Investment Banking & Brokerage

Employee Diversity & Inclusion

Incorporation of Environmental, Social, and Governance Factors in Investment Banking & Brokerage Activities : Environmental, social and governance (ESG) factors may have material impacts on the entities assets and projects across a range of industries to which investment banks provide services or in which they invest.

Business Ethics

Professional Integrity

Systemic Risk Management : Investment Banking & Brokerage entities that fail to manage risks to capital effectively may suffer significant value losses to their financial assets while increasing liabilities.

Employee Incentives & Risk-taking : Variations in employee compensation structures in the Investment Banking & Brokerage industry may incentivise employees to focus on short- or long-term entity performance.

21. Mortgage Finance

Lending Practices : The approach mortgage finance entities take when incentivising employees and how they communicate with customers is important for more than one reason.

Discriminatory Lending : The Mortgage Finance industry aggregates data to determine loan terms and conditions including important provisions such as loan size, interest rate, up-front points or other fees.

Environmental Risk to Mortgaged Properties

22. Security & Commodity Exchanges

Promoting Transparent & Efficient Capital Markets : Security and commodity exchanges have a responsibility to ensure equal access to capital markets for all investors.

Managing Conflicts of Interest

Managing Business Continuity & Technology Risks : Security and commodity exchanges face increased risks and opportunities associated with information technology.

23. Agricultural Products

- Greenhouse Gas Emissions**
- Energy Management**
- Water Management**
- Food Safety**
- Workforce Health & Safety**
- Environmental & Social Impacts of Ingredient Supply Chain**
- GMO Management**
- Ingredient Sourcing**

24. Alcoholic Beverages

- Energy Management**
- Water Management**
- Responsible Drinking & Marketing** : The irresponsible consumption of alcoholic beverages can lead to negative social externalities such as drunk driving, addiction, underage drinking, death and many other public health issues.
- Packaging Lifecycle Management** : Packaging materials represent a significant cost to entities in the Alcoholic Beverages industry.
- Environmental & Social Impacts of Ingredient Supply Chain**
- Ingredient Sourcing**

25. Food Retailers & Distributors

- Fleet Fuel Management**
- Air Emissions from Refrigeration**
- Energy Management**
- Food Waste Management**
- Data Security**
- Food Safety**
- Product Health & Nutrition** : Consumer awareness of food content and nutritional value and their relationship to health, shapes the industry's competitive landscape.
- Product Labelling & Marketing** : Communication with consumers through product labelling and marketing is an important facet of food retail.
- Labour Practices**
- Management of Environmental & Social Impacts in the Supply Chain**

26. Meat, Poultry & Dairy

- Greenhouse Gas Emissions**
- Energy Management**
- Water Management**
- Land Use & Ecological Impacts**
- Food Safety**
- Antibiotic Use in Animal Production** : In livestock production, prevalent use of antibiotics that are also administered to humans may promote the development of antibiotic-resistant strains of bacteria.
- Workforce Health & Safety**
- Animal Care & Welfare** : Entities in the Meat, Poultry & Dairy industry are especially sensitive to changes in the public perception of animal welfare.
- Environmental & Social Impacts of Animal Supply Chain**
- Animal & Feed Sourcing**

27. Non-Alcoholic Beverages

- Fleet Fuel Management**
- Energy Management**
- Water Management**
- Health & Nutrition** : Nutritional and health concerns such as obesity, ingredient safety, nutritional content and adverse health impacts resulting from the consumption of non-alcoholic beverages are important factors in how...
- Product Labelling & Marketing** : Communication with consumers through product labelling and marketing is an important facet of the Non-Alcoholic Beverages industry.

Packaging Lifecycle Management : Packaging materials represent a significant cost to entities in the Non-Alcoholic Beverages industry.

Environmental & Social Impacts of Ingredient Supply Chain

Ingredient Sourcing

28. Processed Foods

Energy Management

Water Management

Food Safety

Health & Nutrition : Nutritional and health concerns such as obesity, ingredient safety and nutritional value are important factors in how entities compete with one another.

Product Labelling & Marketing : Communication with consumers through product labelling and marketing is an important facet of the Processed Foods industry.

Packaging Lifecycle Management : Packaging materials represent a major business cost and contribute to the environmental footprint of entities in the Processed Foods industry.

Environmental & Social Impacts of Ingredient Supply Chain

Ingredient Sourcing

29. Restaurants

Energy Management

Water Management

Food & Packaging Waste Management

Food Safety

Nutritional Content : Public health concerns around obesity have focused on the Restaurants industry.

Labour Practices

Supply Chain Management & Food Sourcing

30. Tobacco

Public Health : Tobacco use can create serious health risks as established by many scientific studies over the past several decades.

Marketing Practices : Tobacco product labelling and marketing is regulated heavily internationally.

Health Care

31. Biotechnology & Pharmaceuticals

Safety of Clinical Trial Participants

Access to Medicines

Affordability & Pricing

Drug Safety

Counterfeit Drugs : Fake or substandard medication presents a significant risk to consumers in all countries.

Ethical Marketing : Biotechnology & Pharmaceuticals entities face challenges associated with the marketing of specific products.

Employee Recruitment, Development & Retention

Supply Chain Management

Business Ethics

32. Drug Retailers

Energy Management in Retail

Data Security & Privacy

Drug Supply Chain Integrity

Management of Controlled Substances : Drug Retailers are distributors and sellers of a wide variety of controlled substances.

Patient Health Outcomes : Drug Retailers and pharmacists play an important role in the health care system, since they provide patients with medications and are often the last health care professionals to interact with patients before...

33. Health Care Delivery

Energy Management

Waste Management

Patient Privacy & Electronic Health Records

Access for Low-Income Patients

Quality of Care & Patient Satisfaction

Management of Controlled Substances : The Health Care Delivery industry is in a unique position with respect to the evolving use of controlled substances and managing the risk of addiction.

Pricing & Billing Transparency : Concern regarding pricing and billing transparency in the Health Care Delivery industry has resulted in increased legal and regulatory scrutiny in some jurisdictions.

Workforce Health & Safety

Employee Recruitment, Development & Retention

Climate Change Impacts on Human Health & Infrastructure

Fraud & Unnecessary Procedures

34. Health Care Distributors

Fleet Fuel Management

Product Safety

Counterfeit Drugs : The World Health Organization (WHO) estimates that counterfeit drugs represent more than 10% of the pharmaceutical supply chain in low- and middle-income countries.

Product Lifecycle Management : Health Care Distributors have a responsibility to reduce the environmental impact of the products that they distribute.

Business Ethics

35. Managed Care

Customer Privacy & Technology Standards

Access to Coverage

Plan Performance

Improved Outcomes : Entities in the Managed Care industry can play a critical role in maintaining and improving the health of enrollees.

Climate Change Impacts on Human Health

36. Medical Equipment & Supplies

Affordability & Pricing

Product Safety

Ethical Marketing : Entities in the Medical Equipment & Supplies industry face legal and regulatory challenges associated with product marketing.

Product Design & Lifecycle Management : Medical equipment and supplies entities face increasing challenges associated with the human and environmental impact of the industry's products.

Supply Chain Management

Business Ethics

Infrastructure

37. Electric Utilities & Power Generators

Greenhouse Gas Emissions & Energy Resource Planning

Air Quality

Water Management

Coal Ash Management

Energy Affordability

Workforce Health & Safety

End-Use Efficiency & Demand : Energy efficiency is a low-lifecycle-cost method to reduce greenhouse gas (GHG) emissions, because less electricity needs to be generated to provide the same end-use energy services.

Nuclear Safety & Emergency Management : Although rare, nuclear accidents can have significant human health and environmental consequences because of their severity.

Grid Resiliency : Electricity is critical for the continued function of most elements of modern life, from medicine to finance, creating a societal reliance on continuous service.

38. Engineering & Construction Services

Environmental Impacts of Project Development

Structural Integrity & Safety

Workforce Health & Safety

Lifecycle Impacts of Buildings & Infrastructure : Buildings and major infrastructure projects are among the largest users of natural resources in the economy; during construction, these materials include iron and steel products, cement, concrete, bricks...

Climate Impacts of Business Mix : Engineering & Construction Services industry clients may be exposed to potentially disruptive climate regulation as well as those that mitigate climate change.

Business Ethics

39. Gas Utilities & Distributors

Energy Affordability

End-Use Efficiency : Natural gas produces fewer greenhouse gas (GHG) emissions than other fossil fuels.

Integrity of Gas Delivery Infrastructure : Operating a vast network of gas pipelines, equipment and storage facilities requires a multifaceted, long-term approach to ensuring infrastructure integrity and managing related risks.

40. Home Builders

Land Use & Ecological Impacts

Workforce Health & Safety

Design for Resource Efficiency : Residential buildings, when occupied, consume significant amounts of energy and water.

Community Impacts of New Developments : Community and urban planning provide home builders with the opportunity to thoughtfully design new residential developments in ways that benefits customers as well as the surrounding community.

Climate Change Adaptation : The impacts of climate change, including extreme weather events and changing climate patterns, may affect the markets entities select to develop homes and residential communities.

41. Real Estate

Energy Management

Water Management

Management of Tenant Sustainability Impacts : Real estate assets generate significant sustainability impacts, including resource consumption (energy and water), waste generation and impacts on occupant health through indoor environmental quality.

Climate Change Adaptation

42. Real Estate Services

Sustainability Services : In the Real Estate Services industry, buildings owned or occupied by clients generally have significant sustainability impacts.

Transparent Information & Management of Conflict of Interest

43. Waste Management

Greenhouse Gas Emissions

Fleet Fuel Management

Air Quality

Management of Leachate & Hazardous Waste

Labour Practices

Workforce Health & Safety

Recycling & Resource Recovery : Recycling, reuse, composting and incineration are general methods of diverting waste from landfills.

44. Water Utilities & Services

Energy Management

Distribution Network Efficiency

Effluent Quality Management

Water Affordability & Access

Drinking Water Quality

End-Use Efficiency : Consumer level water efficiency and conservation—whether a product of government mandates, environmental consciousness or demographic trends—is increasingly important for long-term resource availability and...

Water Supply Resilience

Network Resiliency & Impacts of Climate Change

Renewable Resources & Alternative Energy

45. Biofuels

Air Quality

Water Management in Manufacturing

Lifecycle Emissions Balance : The rapid growth in global biofuels production has been encouraged by government energy policies that seek to reduce net GHG emissions from transportation fuels and dependence on fossil fuels.

Sourcing & Environmental Impacts of Feedstock Production

Management of the Legal & Regulatory Environment : The Biofuels industry is dependent on government policies and regulations that create market demand and incentivise supply with tax breaks and other support for feedstock production.

Operational Safety, Emergency Preparedness & Response : Biofuel production presents operational safety hazards because of the presence of flammable and explosive substances, high temperatures, and pressurised equipment.

46. Forestry Management

Ecosystem Services & Impacts

Rights of Indigenous Peoples

Climate Change Adaptation

47. Fuel Cells & Industrial Batteries

Energy Management

Workforce Health & Safety

Product Efficiency : Both customer demand and regulatory requirements are driving innovation in energy-efficient products with lower environmental impacts and lower total cost of ownership.

Product End-of-life Management : As the rate of adoption of fuel cells and industrial batteries increases and more products reach their end of life, designing products to facilitate end-of-life management and maximise materials efficiency may...

Materials Sourcing

48. Pulp & Paper Products

Greenhouse Gas Emissions

Air Quality

Energy Management

Water Management

Supply Chain Management

49. Solar Technology & Project Developers

Energy Management in Manufacturing

Water Management in Manufacturing

Hazardous Waste Management

Ecological Impacts of Project Development

Management of Energy Infrastructure Integration & Related Regulations : Entities in the industry have faced challenges in establishing solar energy as a cost-competitive means of energy production and GHG reduction, and they have encountered difficulty in capturing a greater...

Product End-of-life Management : Solar panels may contain hazardous substances as well as reusable materials of high economic value.

Materials Sourcing

50. Wind Technology & Project Developers

Workforce Health & Safety

Ecological Impacts of Project Development : Wind farm development involves siting, land acquisition, permitting and engagement with local stakeholders to manage environmental and community impacts.

Materials Sourcing

Materials Efficiency

Resource Transformation

51. Aerospace & Defence

Energy Management

Hazardous Waste Management

Data Security

Product Safety

Fuel Economy & Emissions in Use-phase : Customer preferences and regulatory incentives are increasing the demand for energy-efficient and reduced-emissions products in the Aerospace & Defence industry.

Materials Sourcing

Business Ethics

52. Chemicals

Greenhouse Gas Emissions

Air Quality

Energy Management

Water Management

Hazardous Waste Management

Community Relations

Workforce Health & Safety

Product Design for Use-phase Efficiency : As increasing resource scarcity and regulations encourage greater materials efficiency and lower energy consumption and emissions, the Chemicals industry may benefit from developing products that enhance...

Safety & Environmental Stewardship of Chemicals : Product safety and stewardship is a critical issue for entities in the Chemicals industry.

Genetically Modified Organisms : Some chemical entities produce crop seeds developed using genetically modified organism (GMO) technology.

Management of the Legal & Regulatory Environment : The Chemicals industry faces strict regulation governing air emissions, water discharge, chemical safety and process safety, among other issues.

Operational Safety, Emergency Preparedness & Response : Health, safety and emergency management is a critical issue for entities in the Chemicals industry.

53. Containers & Packaging

Greenhouse Gas Emissions

Air Quality

Energy Management

Water Management

Waste Management

Product Safety

Product Lifecycle Management : Containers and packaging entities face opportunities and challenges associated with the potential environmental impacts of their products throughout their lifecycle.

Supply Chain Management

54. Electrical & Electronic Equipment

Energy Management

Hazardous Waste Management

Product Safety

Product Lifecycle Management : Electrical and electronic equipment entities face increasing challenges and opportunities associated with environmental and social externalities that may stem from the use of their products.

Materials Sourcing

Business Ethics

55. Industrial Machinery & Goods

Energy Management

Workforce Health & Safety

Fuel Economy & Emissions in Use-phase : Many of the Industrial Machinery & Goods industry's products are powered by fossil fuels and release greenhouse gases (GHGs) and other air emissions during use.

Materials Sourcing

Remanufacturing Design & Services

Services

56. Advertising & Marketing

Data Privacy

Advertising Integrity : Entities have a legal responsibility to ensure their products and services advertising is truthful and not deceptive.

Workforce Diversity & Inclusion

57. Casinos & Gaming

Energy Management

Responsible Gaming : Although the main purpose of gambling is entertainment, the industry faces a negative perception often related to pathological gambling.

Smoke-free Casinos

Internal Controls on Money Laundering

58. Education

Data Security

Quality of Education & Gainful Employment : Increasing tuition payments require more students to finance their education with government and private loans.

Marketing & Recruiting Practices : For-profit education entities that grow the number of students that they admit and enrol will increase revenue.

59. Hotels & Lodging

Energy Management

Water Management

Ecological Impacts

Labour Practices

Climate Change Adaptation

60. Leisure Facilities

Energy Management

Customer Safety

Workforce Health & Safety

61. Media & Entertainment

Media Pluralism : Media pluralism, which is diversity in the broadest sense, includes both external and internal pluralism.

Journalistic Integrity & Sponsorship Identification : Audiences rely on journalists for accurate and timely information on current events.

Intellectual Property Protection & Media Piracy

62. Professional & Commercial Services

Data Security

Workforce Diversity & Engagement

Professional Integrity

Technology & Communications

63. Electronic Manufacturing Services & Original Design Manufacturing

Water Management

Waste Management

Labour Practices

Workforce Conditions, Health & Safety

Product Lifecycle Management : Entities in the Electronic Manufacturing Services (EMS) & Original Design Manufacturing (ODM) industry, along with the industry's customers such as hardware entities, face increasing challenges associated with...

Materials Sourcing

64. Hardware

Product Security

Employee Diversity & Inclusion

Product Lifecycle Management : Entities in the Hardware industry face increasing challenges associated with environmental and social externalities attributed to product manufacturing, transport, use and disposal.

Supply Chain Management

Materials Sourcing

65. Internet Media & Services

Environmental Footprint of Hardware Infrastructure
Data Privacy, Advertising Standards & Freedom of Expression
Data Security
Employee Recruitment, Inclusion & Performance
Intellectual Property Protection & Competitive Behaviour

66. Semiconductors

Greenhouse Gas Emissions
Energy Management in Manufacturing
Water Management
Waste Management
Workforce Health & Safety
Recruiting & Managing a Global & Skilled Workforce
Product Lifecycle Management : As an increasing number of devices become connected to each other and to the internet, semiconductor entities face greater demand for products that increase computing power and decrease energy costs.
Materials Sourcing
Intellectual Property Protection & Competitive Behaviour

67. Software & IT Services

Environmental Footprint of Hardware Infrastructure
Data Privacy & Freedom of Expression
Data Security
Recruiting & Managing a Global, Diverse & Skilled Workforce
Intellectual Property Protection & Competitive Behaviour
Managing Systemic Risks from Technology Disruptions : With trends towards increased cloud computing and Software as a Service (SaaS), software and IT service providers must ensure they have robust infrastructure and policies in place to minimise disruptions to...

68. Telecommunication Services

Environmental Footprint of Operations
Data Privacy
Data Security
Product End-of-life Management
Competitive Behaviour & Open Internet
Managing Systemic Risks from Technology Disruptions : Given the systemic importance of telecommunications networks, systemic or economy-wide disruption may result if the Telecommunication Services network infrastructure is unreliable and prone to business...

Transportation

69. Air Freight & Logistics

Greenhouse Gas Emissions
Air Quality
Labour Practices
Workforce Health & Safety
Supply Chain Management
Accident & Safety Management : All modes of transportation pose safety risks.

70. Airlines

Greenhouse Gas Emissions
Labour Practices
Competitive Behaviour
Accident & Safety Management : Air travel accidents may result in significant consequences.

71. Auto Parts

Energy Management
Waste Management
Product Safety

Design for Fuel Efficiency : Automobile manufacturers increasingly are demanding motor parts and components that reduce vehicle fuel consumption.

Materials Sourcing

Materials Efficiency

Competitive Behaviour

72. Automobiles

Product Safety

Labour Practices

Fuel Economy & Use-phase Emissions : Motor vehicle fossil fuel combustion accounts for a significant share of the greenhouse gas (GHG) emissions contributing to global climate change.

Materials Sourcing

Materials Efficiency & Recycling

73. Car Rental & Leasing

Customer Safety

Fleet Fuel Economy & Utilisation : By providing fuel-efficient and alternative fuel vehicles, car rental and leasing entities may improve the environmental sustainability of their operations while also achieving financial benefits.

74. Cruise Lines

Greenhouse Gas Emissions

Air Quality

Discharge Management & Ecological Impacts

Customer Health & Safety

Labour Practices

Workforce Health & Safety

Accident Management : Although cruise ships are one of the safest forms of travel for holidays, the industry competes on customer experience and satisfaction, making safety management a top priority.

75. Marine Transportation

Greenhouse Gas Emissions

Air Quality

Ecological Impacts

Workforce Health & Safety

Business Ethics

Accident & Safety Management : Accidents or leaks involving large vessels can have significant impacts on life, property and the environment.

76. Rail Transportation

Greenhouse Gas Emissions

Air Quality

Workforce Health & Safety

Competitive Behaviour

Accident & Safety Management : Rail accidents and unintended releases of hazardous materials have negative repercussions for the environment and communities along railroad tracks, as well as financial effects on entities themselves.

77. Road Transportation

Greenhouse Gas Emissions

Air Quality

Workforce Conditions, Health & Safety

Accident & Safety Management : Road transportation involves inherent dangers, including accidents resulting from mechanical failure or human error.